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ENGAGEMENT STRUCTURE, AND TERMINAL BLOCK

Abstract

An engagement structure for causing an accommodating member having an opening to engage with an accommodated member which is accommodated in the accommodating member by being inserted into the opening in a rearward direction includes projecting portions and locking holes that engage with one another, and additionally includes dovetail tenons and a dovetail tenon holes that engage with one another, wherein the dovetail tenons engage with the dovetail tenon holes before the projecting portions come into contact with an opening edge of the opening when the accommodated member is being inserted into the accommodating member.

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Background/Summary

TECHNICAL FIELD
[0001] The present invention relates to an engagement structure and a terminal block provided with the engagement structure.
BACKGROUND ART
[0002] JP 2014-135280 A discloses a frame and a mounting portion. The frame has a dovetail block (dovetail tenon). The mounting portion has a dovetail groove (dovetail hole). The dovetail block and the dovetail groove are engaged with each other, whereby the frame and the mounting portion are engaged with each other.
SUMMARY OF THE INVENTION
[0003] An accommodated member is accommodated in an accommodating member through an opening of the accommodating member. A dovetail hole is formed in the accommodating member from an opening end of the opening in the insertion direction of the accommodated member. In addition, the dovetail tenon is formed on the accommodated member in the insertion direction.
[0004] The dovetail tenon and the dovetail hole are engaged with each other when the accommodated member is accommodated in the accommodating member. Thus, the accommodating member and the accommodated member can be engaged with each other while accommodating the accommodated member in the accommodating member.
[0005] Here, a problem arises in that the engagement between the dovetail hole and the dovetail tenon is easily released when the accommodated member moves relative to the accommodating member in the direction opposite to the insertion direction. For example, in the state where the accommodated member is accommodated in the accommodating member, the opening of the accommodating member is directed downward. In this case, the accommodated member is moved relative to the accommodating member in the direction opposite to the insertion direction under the action of gravity. As a result, there is a concern that the engagement between the dovetail hole and the dovetail tenon will be released, and the accommodated member will be dropped from the accommodating member.
[0006] The present invention has the object of solving the aforementioned problem.
[0007] A first aspect of the present invention is an engagement structure for engaging an accommodating member including an opening, with an accommodated member, the accommodated member being accommodated in the accommodating member by being inserted into the opening in an insertion direction, the engagement structure including a latch structure and a dovetail structure, wherein the latch structure includes a locking hole formed in the accommodating member at a position spaced from an opening end of the opening in the insertion direction, and a protruding portion provided on the accommodated member in accordance with a location of the locking hole in the accommodating member and configured to engage with the locking hole so as to restrict relative movement of the accommodated member with respect to the accommodating member in an opposite direction that is opposite to the insertion direction, wherein the dovetail structure includes a dovetail tenon provided at an end of the accommodated member, and a dovetail hole formed in the accommodating member from the opening end in the insertion direction and configured to engage with the dovetail tenon while allowing the dovetail tenon to move to a predetermined

position in the insertion direction, and wherein in a case that the accommodated member is inserted into the accommodating member, the dovetail tenon engages with the dovetail hole before the protruding portion and the opening end come into contact with each other.
[0008] A second aspect of the present invention is a terminal block equipped with a housing including an opening, and a module that is accommodated in the housing by being inserted into the opening in an insertion direction, the terminal block including a locking hole formed in the housing at a position spaced from an opening end of the opening in the insertion direction, a protruding portion provided on the module in accordance with a location of the locking hole in the housing and configured to engage with the locking hole so as to restrict relative movement of the module with respect to the housing in an opposite direction that is opposite to the insertion direction, a dovetail tenon provided at an end of the module, and a dovetail hole formed in the housing from the opening end in the insertion direction and configured to engage with the dovetail tenon while allowing the dovetail tenon to move to a predetermined position in the insertion direction, wherein in a case that the module is inserted into the housing, the dovetail tenon engages with the dovetail hole before the protruding portion and the opening end come into contact with each other.
[0009] According to the present invention, any concern that the accommodated member may fall out of the accommodating member is reduced.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0010] FIG. **1**A is a perspective view of an I/O unit according to an embodiment, and FIG. **1**B is an exploded view of the I/O unit;
[0011] FIG. **2** is an exploded view of a terminal block;
[0012] FIG. **3** is a left side view of the terminal block;
[0013] FIG. **4** is a schematic view showing a state in which a tip of a dovetail tenon has reached an open end of a housing in a module accommodating operation in the housing;
[0014] FIG. **5** is a schematic diagram showing the housing and a module that is further moved in a rearward direction from the state of FIG. **4**;
[0015] FIG. **6**A is a schematic diagram showing the housing and the module that is further moved in the rearward direction from the state of FIG. **5**, and FIG. **6**B is a view of the terminal block shown in FIG. **6**A as viewed from another direction;
[0016] FIG. **7**A is a schematic diagram of a terminal block according to a comparative example, and FIG. **7**B is a view of the terminal block shown in FIG. **7**A as viewed from another direction;
[0017] FIG. **8** is a diagram showing a module according to an Exemplary Modification 1;
[0018] FIG. **9** is a diagram showing a module and a housing according to an Exemplary Modification 2; and
[0019] FIG. **10** is a diagram showing a module and a housing according to an Exemplary Modification 3.
DETAILED DESCRIPTION OF THE INVENTION
Embodiment
[0020] FIG. **1**A is a perspective view of an I/O unit **10** according to an embodiment. FIG. **1**B is an exploded view of the I/O unit **10**.
[0021] The arrows shown in the drawings indicate directions (front, rear, left, right, up, and down) used in the following description. The up-down direction (first direction), the left-right direction (second direction), and the front-rear direction are orthogonal to each other. The left-right direction and the front-rear direction are parallel to, for example, a horizontal plane.
[0022] The I/O unit **10** is provided in a communication system including, for example, a PLC (Programmable Logic Controller). The I/O unit **10** is an electronic device (input/output device)

including a terminal block **12** and a signal processing unit **14**.

[0023] The signal processing unit **14** includes a plurality of contactors **14***a* and a signal processing module **14***b*.

[0024] The plurality of contactors **14***a* are terminal groups disposed on both sides of the signal processing unit **14** in the left-right direction. The plurality of contactors **14***a* disposed on the left side of the signal processing unit **14** are provided to electrically connect to another I/O unit **10** disposed on the left side of the signal processing unit **14**. The plurality of contactors **14***a* disposed on the right side of the signal processing unit **14** are provided to electrically connect to another I/O unit **10** disposed on the right side of the signal processing unit **14**.

[0025] In the present embodiment, the I/O unit **10** is substantially symmetrical in the left-right direction. In view of this, the right side portion of the I/O unit **10** is not shown.

[0026] The signal processing module **14***b* is a module that performs processing required of the I/O unit **10** in the above-described communication system. The signal processing module **14***b* includes an electric circuit such as a control IC (Integrated Circuit). The signal processing module **14***b* is electrically connected to the plurality of contactors **14***a* in the signal processing unit **14**.

[0027] The terminal block **12** is disposed in front of the signal processing unit **14**. The terminal block **12** is mounted on the signal processing unit **14** from the front side of the signal processing unit **14** in the rearward direction. The terminal block **12** includes a plurality of connection ports **16** and an engagement structure **18**.

[0028] Each of the plurality of connection ports **16** is a portion to which an external terminal **20** is connected. The external terminal **20** is a terminal provided in, for example, various actuators used for a machine tool, a robot, etc., and various detectors and the like. A terminal (**401**) for electrical connection with the external terminal **20** is disposed in each of the connection ports **16** (see FIG. **2**). The external terminal **20** is connected to the signal processing module **14***b* through the terminal.

[0029] The engagement structure **18** includes dovetail structures **22** and latch structures **24**. Each of the dovetail structures **22** has a dovetail tenon **26** and a dovetail hole **28** (see FIG. **2**). Each of the latch structures **24** includes a protruding portion **30** and a locking hole **32** (see FIG. **2**).

[0030] The number of the dovetail structures **22** and the latch structures **24** is not particularly limited. The engagement structures **18** shown in FIGS. **1**A (FIG. **1**B) include, for illustration purposes, the plurality of dovetail structures **22** and the plurality of latch structures **24**.

[0031] FIG. **2** is an exploded view of the terminal block **12**.

[0032] The terminal block **12** may be divided into a housing **34** and a module **36**.

[0033] The housing **34** is an accommodating member for accommodating the module **36**. The housing **34** is formed with an opening **38**, a plurality of dovetail holes **28**, and a plurality of locking holes **32**. The module **36** is inserted into the housing **34** through the opening **38**. The opening direction of the opening **38** is a forward direction. Accordingly, an opening end **38***a* of the opening **38** is located at a front end of the housing **34**.

[0034] The module **36** is inserted into the housing **34** from the front side of the housing **34** in the rearward direction. In this case, the insertion direction of the module **36** is the rearward direction.

[0035] Each of the plurality of dovetail holes **28** is a notch formed in the housing **34** from the opening end **38***a* of the opening **38** in the rearward direction. The plurality of dovetail holes **28** are formed in the side portions of the housing **34** in the left-right direction and the up-down direction. In the present embodiment, a case where the plurality of dovetail holes **28** are formed in both sides of the housing **34** in the left-right direction will be described.

[0036] The plurality of locking holes **32** are formed in the side portions of the housing **34** where the plurality of dovetail holes **28** are formed. That is, in the case that the plurality of dovetail holes **28** are provided in a left side portion **34**L and a right side portion **34**R of the housing **34**, the plurality of locking holes **32** are also provided in the left side portion **34**L and the right side portion **34**R of the housing **34**. In the case that the plurality of dovetail holes **28** are arranged in an upper side portion **34**U of the housing **34**, the plurality of locking holes **32** are also arranged in the upper side portion **34**U of the housing **34**. In the case that the plurality of dovetail holes **28** are arranged in a lower side portion **34**B of the housing **34**, the plurality of locking holes **32** are also arranged in the lower side portion **34**B of the housing **34**.

[0037] Each of the plurality of locking holes **32** is disposed in a location of the housing **34** away from the opening end **38***a* in the insertion direction. That is, in the present embodiment, each of the locking holes **32** is disposed rearward of the opening end **38***a*.

[0038] It is preferable for the material of the housing **34** to include a flexible material. The flexible material includes, for example, a resin material. This allows the housing **34** to deform to a certain extent in accordance with the force applied to the housing **34**.

[0039] The module **36** is an accommodated member accommodated in the housing **34**. The module **36** includes a plurality of components **40**.

[0040] The plurality of components **40** include, for example, terminals **401** arranged in the connection ports **16**. However, the plurality of components **40** may further include a release operation element **402** and a light guide element **403**.

[0041] The release operation element **402** is an operation element operated by an operator to release the connection between the connection port **16** and the external terminal **20** connected to the connection port **16**. The release operation element **402** is, for example, a pusher. The terminal block **12** may include a plurality of release operation elements **402**. Accordingly, the plurality of components **40** may include the plurality of release operation elements **402**.

[0042] For example, the light guide element **403** is an element that guides light from a predetermined light source to a lamp provided in the terminal block **12** in order to turn on the lamp. The light guide element **403** is, for example, a light pipe. The terminal block **12** may include a plurality of light guide elements **403**. Accordingly, the plurality of components **40** may include a plurality of light guide elements **403**.

[0043] The module **36** is formed with a plurality of dovetail tenons **26** and the plurality of protruding portions **30**.

[0044] The plurality of dovetail tenons **26** are provided on the sides of the module **36** in accordance with the locations of the plurality of dovetail holes **28** in the housing **34**. That is, in the case that the plurality of dovetail holes **28** are arranged in the left side portion **34**L and the right side portion **34**R of the housing **34**, the plurality of dovetail tenons **26** are arranged in a left side portion **36**L and a right side portion **36**R of the module **36**. In the case that the plurality of dovetail holes **28** are arranged in the upper side portion **34**U of the housing **34**, the plurality of dovetail tenons **26** are also arranged in an upper side portion **36**U of the module **36**. In the case that the plurality of dovetail holes **28** are arranged in the lower side portion **34**B of the housing **34**, the plurality of dovetail tenons **26** are also arranged in the lower side portion **36**B of the module **36**.

[0045] Each of the plurality of dovetail tenons **26** extends in the insertion direction from an end of the module **36** that faces opposite to the insertion direction. That is, in the present embodiment, each of the dovetail tenons **26** extends in the rearward direction from an end **36***a* of the module **36** facing in the forward direction.

[0046] The plurality of protruding portions **30** are provided on the sides of the module **36** in accordance with the locations of the plurality of locking holes **32** in the housing **34**. That is, in the case that the plurality of locking holes **32** are provided in the left side portion **34**L and the right side portion **34**R of the housing **34**, the plurality of protruding portions **30** are provided in the left side portion **36**L and the right side portion **36**R of the module **36**. In the case that the plurality of locking holes **32** are arranged in the upper side portion **34**U of the housing **34**, the plurality of protruding portions **30** are also arranged in the upper side portion **36**U of the module **36**. In the case that the plurality of locking holes **32** are arranged in the lower side portion **34**B of the housing **34**, the plurality of protruding portions **30** are also arranged in the lower side portion **36**B of the module **36**.

[0047] FIG. **3** is a left side view of the terminal block **12**.

[0048] The plurality of dovetail tenons **26** are provided corresponding to the dovetail holes **28** that are different from each other. Each of the dovetail tenons **26** engages the corresponding dovetail hole **28** in the case that the module **36** is accommodated in the housing **34**.
[0049] Since the respective dovetail holes **28** are long in the insertion direction, each of the dovetail holes **28** allows the dovetail tenon **26** to move to a certain extent in the insertion direction while engaging with the corresponding dovetail tenon **26**. Each of the dovetail holes **28** allows the corresponding dovetail tenon **26** to move to a predetermined position in the insertion direction. For example, the predetermined positions are positions where both side portions (**36**L, **36**R) of the module **36** in the left-right direction are completely accommodated in the housing **34**.
[0050] The engagement of the dovetail tenons **26** and the dovetail holes **28** restricts the relative movement of the module **36** with respect to the housing **34** in a direction perpendicular to the insertion direction. For example, in the case of the present embodiment, the engagement between the dovetail tenons **26** and the dovetail holes **28** restricts the relative movement of the module **36** with respect to the housing **34** in the up-down direction and the left-right direction.
[0051] Further, the engagement between the dovetail tenons **26** and the dovetail holes **28** restricts the side portion of the housing **34** where the dovetail hole **28** is formed from being bent. For example, in the case of the present embodiment, the engagement between the dovetail tenons **26** of the left side portion **36**L of the module **36** and the dovetail holes **28** of the left side portion **34**L of the housing **34** restricts the left side portion **34**L of the housing **34** from being bent. For example, in the case of the present embodiment, the engagement between the dovetail tenons **26** of the right side portion **36**R of the module **36** and the dovetail holes **28** of the right side portion **34**R of the housing **34** restricts the right side portion **34**R of the housing **34** from being bent.
[0052] The plurality of protruding portions **30** are provided corresponding to the locking holes **32** that are different from each other. Each of the protruding portions **30** engages the corresponding locking hole **32** in the case that the module **36** is accommodated in the housing **34**.
[0053] The engagement of the protruding portions **30** with the locking holes **32** restricts the relative movement of the module **36** with respect to the housing **34** in the insertion direction and the direction opposite to the insertion direction. For example, in the case of the present embodiment, the engagement between the protruding portions **30** and the locking holes **32** restricts the relative movement of the module **36** with respect to the housing **34** in the front-rear direction.
[0054] The engagement of the protruding portions **30** with the locking holes **32** restricts the relative movement of the module **36** with respect to the housing **34**, in the extending direction of the side portion of the housing **34** where the locking holes **32** are formed. For example, in the case of the present embodiment, the left and right side portions (**34**L, **34**R) of the housing **34** extend in the up-down direction and the front-rear direction. Accordingly, the engagement between the protruding portions **30** and the locking holes **32** restricts the relative movement of the module **36** with respect to the housing **34** in the up-down direction and the front-rear direction.
[0055] In the above-mentioned terminal block **12**, the engagement between the protruding portions **30** and the locking holes **32** restricts the relative movement of the module **36** with respect to the housing **34** in the front-rear direction. That is, the latch structure **24** (the protruding portion **30** and the locking hole **32**) can restrict the relative movement of the module **36** with respect to the housing **34**, in the insertion direction in which the movement of the dovetail structure **22** (the dovetail tenon **26** and the dovetail hole **28**) cannot be restricted. This prevents, for example, the module **36** from inadvertently falling out of the housing **34** through the opening **38**.
[0056] The engagement of the dovetail tenons **26** and the dovetail holes **28** restricts the relative movement of the module **36** with respect to the housing **34** in the up-down direction. Further, the engagement between the protruding portions **30** and the locking holes **32** more strongly restricts the relative movement of the module **36** with respect to the housing **34** in the up-down direction.
[0057] Further, the engagement between the dovetail tenons **26** and the dovetail holes **28** restricts the relative movement of the module **36** with respect to the housing **34** in the left-right direction.

That is, the dovetail structure **22** (the dovetail tenon **26** and the dovetail hole **28**) can restrict the relative movement of the module **36** with respect to the housing **34**, in the direction in which the latch structure **24** (the protruding portion **30** and the locking hole **32**) cannot restrict the movement. By restricting the relative movement of the module **36** with respect to the housing **34** in the left-right direction, for example, the engagement between the protruding portion **30** and the locking hole **32** is prevented from being released inadvertently due to the relative movement.

[0058] In addition, the engagement between the dovetail tenons **26** and the dovetail holes **28** restricts the housing **34** from being bent in the left-right direction. This prevents the engagement between the protruding portions **30** and the locking holes **32** from being released inadvertently due to the bending.

[0059] The terminal block **12** can restrict the relative movement of the module **36** with respect to the housing **34** in all directions of the front-rear direction, the left-right direction, and the up-down direction. In addition, the state in which the module **36** is accommodated in the housing **34** can be maintained more reliably.

[0060] Each of the dovetail tenons **26** has a tip **26***a* in the insertion direction. Each of the protruding portions **30** has a tip **30***a* in the insertion direction. In this case, with respect to the dovetail tenons **26** and the protruding portions **30** disposed on the same side of the module **36**, it is preferable for the tips **26***a* of the dovetail tenons **26** to be located further in the insertion direction relative to the tips **30***a* of the protruding portions **30**. That is, in the case of the present embodiment, it is preferable for each of the tips **26***a* of the dovetail tenons **26** to be located rearward of each of the tips **30***a* of the plurality of protruding portions **30** in each of the left side portion **36**L and the right side portion **36**R of the module **36** (see also FIG. **3**).

[0061] The process of accommodating operation the module **36** in the housing **34** will be described below, including the reason for the above positional relationship. In the following description, the tips **26***a* of the dovetail tenons **26** are located rearward of the tips **30***a* of the protruding portions **30**.

[0062] FIG. **4** is a schematic diagram showing a state in which the tips **26***a* of the dovetail tenons **26** have reached the opening end **38***a* of the housing **34** in the accommodating operation of the module **36** in the housing **34**.

[0063] As previously described, the module **36** is accommodated in the housing **34** through the opening **38** of the housing **34**. By moving the module **36** from the front to the rear of the housing **34**, the tip **26***a* of each of the plurality of dovetail tenons **26** reaches the opening end **38***a* of the housing **34**. In this state, the tip **30***a* of each of the plurality of protruding portions **30** does not reach the opening end **38***a*.

[0064] FIG. **5** is a schematic diagram showing the housing **34** and the module **36** further moved in the rearward direction from the state of FIG. **4**.

[0065] In FIG. **5**, the tip **30***a* of each of the plurality of protruding portions **30** reaches the opening end **38***a*. In this state, the end portion of each of the plurality of dovetail tenons **26** is engaged with the dovetail hole **28**.

[0066] FIG. **6**A is a schematic diagram showing the housing **34** and the module **36** further moved in the rearward direction from the state of FIG. **5**. FIG. **6**B is a view of the terminal block **12** shown in FIG. **6**A as viewed from another direction.

[0067] FIG. **6**B is a view of the terminal block **12** shown in FIG. **6**A as seen from the front. However, in order to avoid the complicated illustration, the plurality of connection ports **16**, the components **40**, and the like are not omitted.

[0068] After the tip **30***a* of each of the plurality of protruding portions **30** reaches the opening end **38***a*, the tip **30***a* of each protruding portion **30** enters the housing **34** by moving the module **36** further in the rearward direction.

[0069] Each of the plurality of protruding portions **30** that have entered the housing **34** presses the housing **34** in the protruding direction in which the protruding portions **30** each protrude from the inside of the housing **34**. For example, in the case of the present embodiment, the plurality of protruding portions **30** arranged on the left side portion **36**L of the module **36** protrude in the left direction. Also, in the case of the present embodiment, the plurality of protruding portions **30** arranged on the right side portion **36**R of the module **36** protrude in the right direction.

[0070] Therefore, the housing **34** receives forces in directions in which the housing **34** is pushed and spread in the left-right direction from the plurality of protruding portions **30**. The housing **34** deforms in accordance with the forces received from the plurality of protruding portions **30**. However, as described above, the engagement of the dovetail tenons **26** and the dovetail holes **28** restricts both of the side portions (**34**L, **34**R) of the housing **34** in the left-right direction from being bent. Therefore, the housing **34** does not bend greatly in the left-right direction.

[0071] Before and after the housing **34** is deformed in accordance with the shape of the protruding portions **30**, each of the dovetail tenons **26** continues to be moved in the rearward direction along the corresponding dovetail hole **28**.

[0072] When the module **36** is further moved in the rearward direction from the state shown in FIGS. **6**A and **6**B, each of the plurality of protruding portions **30** reaches the corresponding locking hole **32**. As a result, the plurality of protruding portions **30** and the plurality of locking holes **32** are engaged with each other. That is, the terminal block **12** in the state shown in FIG. **3** is obtained.

[0073] FIG. **7**A is a schematic diagram of a terminal block **100** according to a comparative example.

[0074] The terminal block **100** includes a housing **102** and a module **104**.

[0075] The housing **102** includes an opening **105**, a plurality of dovetail holes **106**, and a plurality of locking holes **108**. The opening direction of the opening **105** is a forward direction. An opening end **105***a* of the opening **105** is located at a front end of the housing **102**. The housing **102** has flexibility.

[0076] The module **104** includes a plurality of dovetail tenons **110** and a plurality of protruding portions **112**. The module **104** is accommodated in the housing **102** from the front side of the housing **102** in the rearward direction.

[0077] The plurality of dovetail tenons **110** are provided corresponding to the dovetail holes **106** that are different from each other. The plurality of protruding portions **112** are arranged corresponding to the locking holes **108** that are different from each other.

[0078] In this case, tips **110***a* of the respective dovetail tenons **110** in the rearward direction are located forward of tips **112***a* of the protruding portions **112** in the rearward direction. Accordingly, in the case that the module **104** is accommodated in the housing **102**, the protruding portions **112** enter the housing **102** before the dovetail tenons **110** and the dovetail holes **106** are engaged with each other.

[0079] FIG. **7**B is a view of the terminal block **100** shown in FIG. **7**A as viewed from another direction.

[0080] As the protruding portions **112** enter the housing **102**, the housing **102** deforms. Here, the housing **102** is deformed before the dovetail holes **106** and the locking holes **108** are engaged with each other. Therefore, the position of the edge portion of each of the dovetail holes **106** may be greatly changed in accordance with the deformation of the housing **102**.

[0081] For instance, in the example of FIG. **7**B, the edge portions of the plurality of dovetail holes **106** arranged in a left side portion **102**L of the housing **102** are significantly shifted leftward relative to the dovetail tenons **110**. In addition, the edge portions of the plurality of dovetail holes **106** arranged in a right side portion **102**R of the housing **102** are significantly shifted rightward relative to the dovetail tenons **110**. As a result, in the comparative example, it is difficult to engage the plurality of dovetail holes **106** with the plurality of dovetail tenons **110**.

[0082] In this regard, according to the present embodiment, with respect to the dovetail tenons **26** and the protruding portions **30** disposed on the same side of the module **36**, the tips **26***a* of the dovetail tenons **26** are located further in the insertion direction relative to the tips **30***a* of the protruding portions **30**. As a result, the dovetail tenons **26** and the dovetail holes **28** are engaged with each other before the protruding portions **30** deform the housing **34**. That is, the engagement between the dovetail tenons **26** and the dovetail holes **28** is not hindered by the latch structure **24**. Therefore, the operator can easily assemble the terminal block **12** that can more reliably maintain the state in which the module **36** is accommodated in the housing **34**.

Exemplary Modifications

[0083] Hereinafter, a description will be given concerning exemplary modifications of the above-described embodiment. However, in the description that follows, explanations that overlap with those of the above-described embodiment will be omitted insofar as possible. Unless otherwise specified, the same reference numerals as in the above-described embodiment are used in referring to the elements that have already been described in the embodiment.

(Exemplary Modification 1)

[0084] FIG. **8** is a diagram showing a module **361** (**36**) according to an Exemplary Modification 1. In order to simplify the description, the illustration of portions that are not mentioned in the following description is omitted insofar as possible.

[0085] The shape of the protruding portions **30** provided in the module **36** may be modified. That is, the module **361** includes a plurality of protruding portions **301** (**30**). Each of the protruding portions **301** is inclined so that the thickness thereof in the protruding direction becomes smaller in the insertion direction. For example, the module **361** is accommodated in the housing **34** in the rearward direction. In this case, each of the protruding portions **301** is inclined so that the thickness thereof in the protruding direction becomes smaller in the rearward direction.

[0086] By inclining the respective protruding portions **301** as described above, the protruding portions **301** can be smoothly inserted into the housing **34** in the case that the module **361** is accommodated in the housing **34**.

[0087] In addition, in a range where a portion of each of the protruding portions **301** abutting against the opening end **38***a* is located forward of the tip **26***a* of each of the dovetail tenons **26**, the tip **30***a* of each of the protruding portions **301** may be located rearward of the tip **26***a* of each of the dovetail tenons **26**. In addition, within the same range, the tip **30***a* of each of the protruding portions **301** and the tip **26***a* of each of the dovetail tenons **26** may be disposed at substantially the same position in the front-rear direction.

(Exemplary Modification 2)

[0088] FIG. **9** is a diagram showing a module **362** (**36**) and a housing **342** (**34**) according to an Exemplary Modification 2. FIG. **9** shows a part of a left side portion **36**L of the module **362** and a part of a left side portion **34**L of the housing **342**.

[0089] The module **362** includes the constituent elements of the module **36** (see the embodiment). However, a dovetail tenon **26** and protruding portions **30** of the module **362** are connected to each other. A plurality of protruding portions **30** may be connected to a single dovetail tenon **26**. For instance, in the example shown in FIG. **9**, a plurality of protruding portions **30** are arranged so as to sandwich a single dovetail tenon **26** in a direction perpendicular to the insertion direction.

[0090] The housing **342** includes the constituent elements of the housing **34** (see the embodiment). The dovetail hole **28** and the locking holes **32** of the housing **342** are arranged in accordance with the locations of the dovetail tenon **26** and the protruding portions **30** of the module **362**. For example, the dovetail hole **28** and the locking holes **32** of the housing **342** are connected to each other.

[0091] In the case that the module **362** is provided with the plurality of protruding portions **30**, the plurality of protruding portions **30** may include both the protruding portions **30** that are connected to the dovetail tenon **26** and the protruding portions **30** that are not connected to the dovetail tenon **26**.

(Exemplary Modification 3)

[0092] FIG. **10** is a diagram showing a module **363** (**36**) and a housing **343** (**34**) according to an Exemplary Modification 3. FIG. **10** shows a part of a left side portion **36**L of the module **363** and a

part of a left side portion **34**L of the housing **343**.
[0093] The module **363** includes the constituent elements of the module **36** (see the embodiment). The module **363** further includes a projecting portion **42**. The module **363** may include a plurality of projecting portions **42**.
[0094] The protruding direction of the projecting portion **42** coincides with the protruding direction of the protruding portion **30**. Accordingly, in the example of FIG. **10**, the projecting portion **42** protrudes in the left direction.
[0095] The projecting portion **42** is disposed on the same side of the module **363** as the dovetail tenon **26**. For example, in the illustration of FIG. **10**, the dovetail tenon **26** is at least disposed on a left side portion **36**L of the module **363**. Thus, the projecting portion **42** is disposed at least on the left side portion **36**L of the module **363**.
[0096] The projecting portion **42** is connected to the side portion of the dovetail tenon **26** in the direction perpendicular to the insertion direction, which is the extending direction of the side portion of the module **363** on which the dovetail tenon **26** is disposed. For instance, in the example of FIG. **10**, the left side portion **36**L of the module **363** extends in the front-rear direction and the up-down direction. The insertion direction is the rearward direction. The up-down direction among the front-rear direction and the up-down direction, is perpendicular to the rearward direction. In this case, the projecting portion **42** is connected to a lower side portion **26***b* or an upper side portion **26***c* of the dovetail tenon **26**. The projecting portion **42** of FIG. **10** is connected to the lower side portion **26***b*.
[0097] The projecting portion **42** has an inclined surface **42***a*. The inclined surface **42***a* is a surface of the projecting portion **42** in the direction opposite to the insertion direction. Accordingly, in the example of FIG. **10**, the inclined surface **42***a* is a frontward surface of the projecting portion **42**. The inclined surface **42***a* is inclined from the dovetail tenon **26** in the rearward direction (insertion direction).
[0098] The housing **343** includes the constituent elements of the housing **34** (see the embodiment). The housing **343** is further formed with a notch **44** for engaging with the projecting portion **42**.
[0099] The notch **44** has an edge portion **44***a* in the direction opposite to the insertion direction. In the example of FIG. **10**, the edge portion **44***a* is a frontward edge portion of the notch **44**. The edge portion **44***a* is preferably parallel to the inclined surface **42***a* in the case that the module **363** is inserted into the housing **343**.
[0100] According to the present modification, in the case that the module **363** is accommodated in the housing **343**, the inclined surface **42***a* of the projecting portion **42** slides in the insertion direction (rearward direction) along the edge portion **44***a* of the notch **44**. Therefore, according to the present modification, the module **363** can be smoothly accommodated in the housing **343**.
[0101] In the case that the module **363** is provided with a plurality of dovetail tenons **26**, the module **363** may be provided with a plurality of projecting portions **42** corresponding to the plurality of dovetail tenons **26**, respectively. However, the projecting portion **42** corresponding to at least one of the plurality of dovetail tenons **26** may be omitted.
(Exemplary Modification 4)
[0102] The engagement structure **18** according to the present embodiment may be applied to members other than the terminal block **12**. In short, the engagement structure **18** may be applied to other than the terminal block **12**, within a range in which the accommodating member (**34**) including the opening (**38**) and the accommodated member (**36**) accommodated in the accommodating member (**34**) are engaged with each other by the accommodated member (**36**) being inserted into the opening (**38**) in the insertion direction.
(Combination of Plural Exemplary Modifications)
[0103] The above-described respective exemplary modifications may be appropriately combined within a range in which no inconsistencies occur.
[0104] For example, the protruding portion **30** may be connected to one side of the dovetail tenon

26 in the up-down direction (Exemplary Modification 2), and the projecting portion **42** may be connected to another side of the dovetail tenon **26** in the up-down direction (Exemplary Modification 3).

Invention Obtained from the Embodiment

[0105] Invention that can be grasped from the above-described embodiment and the exemplary modifications thereof will be described below.

<First Aspect of Invention>

[0106] A first aspect of the invention is the engagement structure (**18**) for engaging the accommodating member (**34**) including the opening (**38**), with the accommodated member (**36**), the accommodated member (**36**) being accommodated in the accommodating member by being inserted into the opening in the insertion direction, the engagement structure including the latch structure (**24**) and the dovetail structure (**22**), wherein the latch structure includes the locking hole (**32**) formed in the accommodating member at the position spaced from the opening end (**38***a*) of the opening in the insertion direction, and the protruding portion (**30**) provided on the accommodated member in accordance with the location of the locking hole in the accommodating member and configured to engage with the locking hole so as to restrict the relative movement of the accommodated member with respect to the accommodating member in the opposite direction that is opposite to the insertion direction, wherein the dovetail structure includes the dovetail tenon (**26**) provided at the end of the accommodated member, and the dovetail hole (**28**) formed in the accommodating member from the opening end in the insertion direction and configured to engage with the dovetail tenon while allowing the dovetail tenon to move to the predetermined position in the insertion direction, and wherein in the case that the accommodated member is inserted into the accommodating member, the dovetail tenon engages with the dovetail hole before the protruding portion and the opening end come into contact with each other.

[0107] In accordance with such features, any concern that the accommodated member may fall out of the accommodating member is reduced.

[0108] The tip (**26***a*) of the dovetail tenon in the insertion direction may be located further in the insertion direction relative to the tip (**30***a*) of the protruding portion in the insertion direction. In accordance with such features, the engagement on the dovetail structure side is started before the engagement on the latch structure side.

[0109] The protruding portion may be inclined in the manner so that the thickness of the protruding portion in the protruding direction is reduced in the insertion direction, and the tip (**26***a*) of the dovetail tenon in the insertion direction may be located at the same position with respect to the insertion direction as the tip (**30***a*) of the protruding portion in the insertion direction, or located further in the opposite direction relative to the tip of the protruding portion. In accordance with such features, the engagement on the dovetail structure side is started before the engagement on the latch structure side, and the engagement on the latch structure side is smoothly made.

[0110] The protruding portion and the dovetail tenon may be connected. In accordance with such features, any concern that the accommodated member may fall out of the accommodating member is reduced.

[0111] Two of the protruding portions may be provided so as to sandwich the dovetail tenon. In accordance with such features, any concern that the accommodated member may fall out of the accommodating member is reduced.

[0112] The engagement structure may further include the projecting portion (**42**) provided on the accommodated member and connected to the side portion (**26***b*) of the dovetail tenon in the direction perpendicular to the insertion direction, and the notch (**44**) provided in the accommodating member and configured to engage with the projecting portion, wherein the projecting portion may include the inclined surface (**42***a*) on the side in the opposite direction, the inclined surface being inclined in the insertion direction from the dovetail tenon, and the notch may include the edge portion (**44***a*) on the side in the opposite direction, the edge portion being inclined in the insertion direction from the dovetail hole. In accordance with such features, the accommodation operation can be facilitated.

<Second Aspect of Invention>

[0113] The terminal block (**12**) is equipped with the housing (**34**) including the opening (**38**), and the module (**36**) that is accommodated in the housing by being inserted into the opening in the insertion direction, the terminal block including the locking hole (**32**) formed in the housing at the position spaced from the opening end (**38***a*) of the opening in the insertion direction, the protruding portion (**30**) provided on the module in accordance with the location of the locking hole in the housing and configured to engage with the locking hole so as to restrict relative movement of the module with respect to the housing in the opposite direction that is opposite to the insertion direction, the dovetail tenon (**26**) provided at the end of the module, and the dovetail hole (**28**) formed in the housing from the opening end in the insertion direction and configured to engage with the dovetail tenon while allowing the dovetail tenon to move to the predetermined position in the insertion direction, wherein in the case that the module is inserted into the housing, the dovetail tenon engages with the dovetail hole before the protruding portion and the opening end come into contact with each other.

[0114] In accordance with such features, any concern that the module may fall out of the housing is reduced.

Reference Signs List

[0115] **10**: I/O unit [0116] **12**: terminal block [0117] **18**: engagement structure [0118] **20**: external terminal [0119] **22**: dovetail structure [0120] **24**: latch structure [0121] **26**: dovetail tenon [0122] **26***a*: tip of dovetail tenon (tip) [0123] **26***b*: lower side portion of the dovetail tenon (side portion) [0124] **26***c*: upper side portion of dovetail tenon (side portion) [0125] **28**: dovetail hole [0126] **30**, **301**: protruding portion [0127] **30***a*: tip of the protruding portion (tip) [0128] **32**: locking hole [0129] **34**, **342**, **343**: housing [0130] **36**, **361** to **363**: module [0131] **38**: opening [0132] **42**: projecting portion [0133] **42***a*: inclined surface of the projecting portion (inclined surface) [0134] **44**: notch [0135] **44***a*: edge portion of notch (edge portion) [0136] **401**: terminal

Claims

1. An engagement structure for engaging an accommodating member including an opening, with an accommodated member, the accommodated member being accommodated in the accommodating member by being inserted into the opening in an insertion direction, the engagement structure comprising a latch structure and a dovetail structure, wherein the latch structure includes: a locking hole formed in the accommodating member at a position spaced from an opening end of the opening in the insertion direction; and a protruding portion provided on the accommodated member in accordance with a location of the locking hole in the accommodating member and configured to engage with the locking hole so as to restrict relative movement of the accommodated member with respect to the accommodating member in an opposite direction that is opposite to the insertion direction, wherein the dovetail structure includes: a dovetail tenon provided at an end of the accommodated member; and a dovetail hole formed in the accommodating member from the opening end in the insertion direction and configured to engage with the dovetail tenon while allowing the dovetail tenon to move to a predetermined position in the insertion direction, and wherein in a case that the accommodated member is inserted into the accommodating member, the dovetail tenon engages with the dovetail hole before the protruding portion and the opening end come into contact with each other.

2. The engagement structure according to claim 1, wherein a tip of the dovetail tenon in the insertion direction is located further in the insertion direction relative to a tip of the protruding portion in the insertion direction.

3. The engagement structure according to claim 1, wherein the protruding portion is inclined in a manner so that a thickness of the protruding portion in a protruding direction is reduced in the insertion direction, and a tip of the dovetail tenon in the insertion direction is located at a same position with respect to the insertion direction as a tip of the protruding portion in the insertion direction, or located further in the opposite direction relative to the tip of the protruding portion.

4. The engagement structure according to claim 1, wherein the protruding portion and the dovetail tenon are connected.

5. The engagement structure according to claim 1, wherein two of the protruding portions are provided so as to sandwich the dovetail tenon.

6. The engagement structure according to claim 1, further comprising: a projecting portion provided on the accommodated member and connected to a side portion of the dovetail tenon in a direction perpendicular to the insertion direction; and a notch provided in the accommodating member and configured to engage with the projecting portion, wherein the projecting portion includes an inclined surface on a side in the opposite direction, the inclined surface being inclined in the insertion direction from the dovetail tenon, and the notch includes an edge portion on a side in the opposite direction, the edge portion being inclined in the insertion direction from the dovetail hole.

7. A terminal block equipped with a housing including an opening, and a module that is accommodated in the housing by being inserted into the opening in an insertion direction, the terminal block comprising: a locking hole formed in the housing at a position spaced from an opening end of the opening in the insertion direction; a protruding portion provided on the module in accordance with a location of the locking hole in the housing and configured to engage with the locking hole so as to restrict relative movement of the module with respect to the housing in an opposite direction that is opposite to the insertion direction; a dovetail tenon provided at an end of the module; and a dovetail hole formed in the housing from the opening end in the insertion direction and configured to engage with the dovetail tenon while allowing the dovetail tenon to move to a predetermined position in the insertion direction, wherein in a case that the module is inserted into the housing, the dovetail tenon engages with the dovetail hole before the protruding portion and the opening end come into contact with each other.